

8-Week Course Schedule

From Python basics to a deployed AI application — built in 8 weeks.

Weeks 1–2

Phase 1
Foundations

8

Weeks

Weeks 3–4

Phase 2
APIs & Tools

16

Sessions

Weeks 5–6

Phase 3
RAG Systems

8

Deliverables

Weeks 7–8

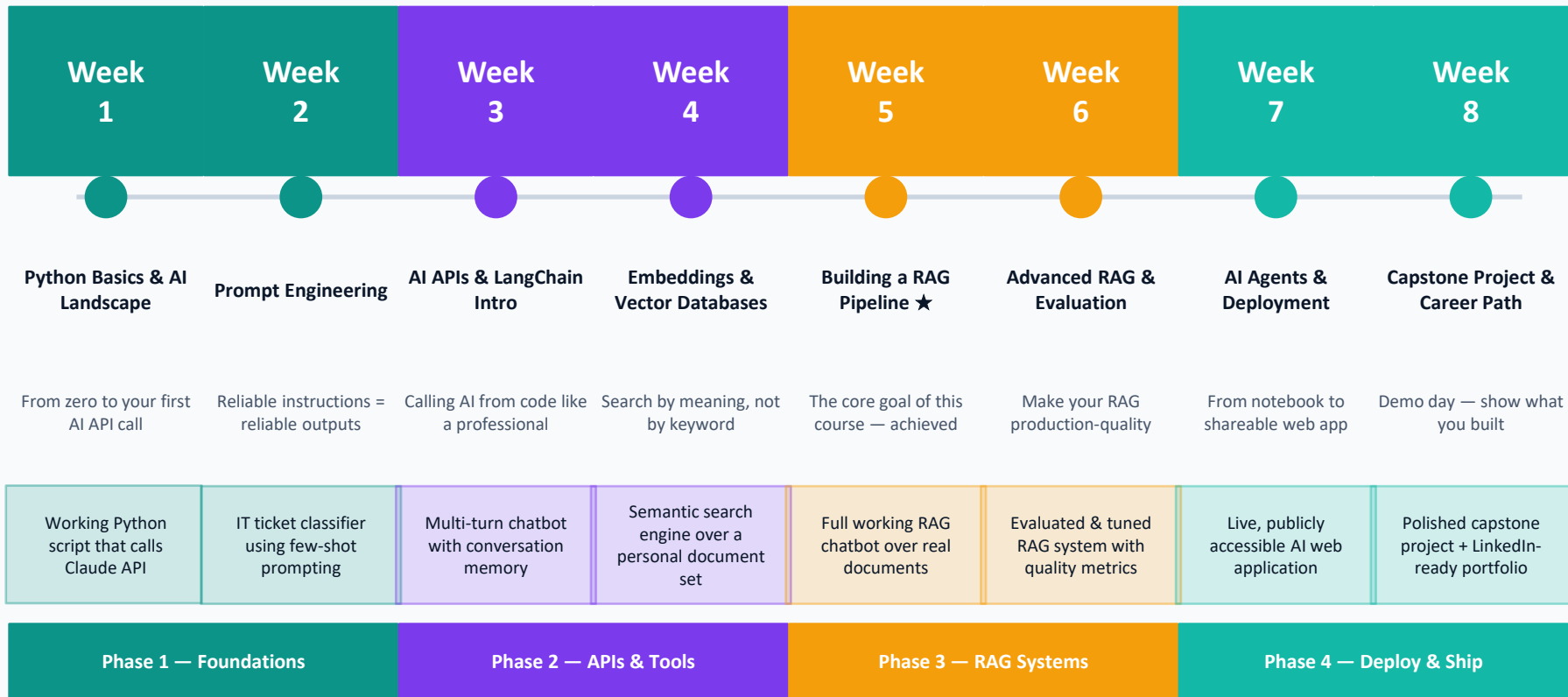
Phase 4
Deploy & Ship

1

Capstone App

8-Week Learning Journey

Application-first. Confidence-first. Foundations progressively.



1

WEEK

Phase 1
Foundations

 Layer 1

HOMEWORK

Use AI for one real work task.
Write 3 sentences on what
worked.

DELIVERABLE

Working Python script that
calls Claude API

Phase 1 — Foundations

Python Basics & AI Landscape

From zero to your first AI API call

✂ Hands-On Lab

Session 1

Python Crash Course

- Variables, strings, lists, dicts
- Loops, conditions, functions
- File I/O & error handling
- Mini-project: Log file analyser

💡 Concept + Code

Session 2

AI Landscape & First API Call

- What is AI, ML, LLMs, GenAI
- The AI family tree (visual)
- Demo: HuggingFace, Ollama, ChatGPT & Claude
- Your first Anthropic API call

2

WEEK

Phase 1
Foundations

Layer 1

HOMEWORK

Build 3 prompts for real tasks
in your current role.

DELIVERABLE

Notebook with examples of
each

Phase 1 — Foundations

Prompt Engineering

Reliable instructions = reliable outputs



Concept + Code

Session 1

Prompt Patterns

- Zero-shot prompting
- Few-shot prompting with examples
- System prompts & persona setting
- Prompt structure best practices



Hands-On Lab

Session 2

Applied Prompt Engineering

- Chain-of-thought prompting
- Output format control (JSON, lists)
- IT-specific use cases
- Lab: usacase

3

WEEK

Phase 2
AI APIs & Tools

 Layer 1

HOMEWORK

Build a multi-turn chatbot on a topic from your work domain.

DELIVERABLE

Chatbot with conversation memory

Phase 2 — AI APIs & Tools

AI APIs & LangChain Intro

Calling AI from code like a professional



Concept + Code

Session 1

Working with AI APIs

- API authentication & key management
- Request/response structure
- Streaming responses
- Multi-turn conversations (chat history)



Hands-On Lab

Session 2

LangChain Fundamentals

- What is LangChain & why use it
- Chains: prompt → model → output
- LangChain vs raw API calls
- Lab: Build a document Q&A chain

4

WEEK

Phase 2
AI APIs & Tools

 Layer 1 + Layer 2

HOMEWORK

Index 10 documents from
your own domain into
ChromaDB.

DELIVERABLE

Semantic search engine over
a personal document set

Phase 2 — AI APIs & Tools

Embeddings & Vector Databases

Search by meaning, not by keyword

💡 Concept + Code

Session 1

Embeddings — Visual Intuition

- What embeddings are (no maths)
- Semantic similarity vs keyword search
- Cosine similarity explained simply
- Hands-on: build semantic search from scratch

✂ Hands-On Lab

Session 2

ChromaDB & Vector Stores

- Vector databases explained
- ChromaDB setup & collections
- Storing documents with metadata
- Lab: Index your IT runbooks & search them

5

WEEK

Phase 3
RAG Systems

 Layer 1 + Layer 2

HOMEWORK

Build a RAG chatbot over at least 5 documents from your work.

DELIVERABLE

🚀 Full working RAG chatbot over real documents

Phase 3 — RAG Systems

Building a RAG Pipeline ★

The core goal of this course — achieved

💡 Concept + Code

Session 1

RAG Architecture

- How RAG works: Retrieve → Augment → Generate
- Chunking strategies & trade-offs
- Context injection into prompts
- Why RAG beats plain prompting

✂ Hands-On Lab

Session 2

Build Your First RAG App

- End-to-end RAG pipeline in Python
- Ask questions over PDF documents
- Source citation & grounding
- Lab: 'Ask my company docs' chatbot

6

WEEK

Phase 3
RAG Systems

Layer 2

HOMework

Run evaluation on your RAG bot. Identify top 2 failure modes.

DELIVERABLE

Evaluated & tuned RAG system with quality metrics

Phase 3 — RAG Systems

Advanced RAG & Evaluation

Make your RAG production-quality



Concept + Code

Session 1

Advanced RAG Techniques

- Re-ranking retrieved results
- Hybrid search (semantic + keyword)
- Query rewriting & expansion
- Handling large documents



Hands-On Lab

Session 2

RAG Evaluation

- How to measure RAG quality
- LLM-as-judge evaluation pattern
- Hallucination detection
- Lab: Evaluate and tune your Week 5 RAG bot

7

WEEK

Phase 4
Deploy & Ship

Layer 1

HOMEWORK

Deploy your RAG app and
share the URL in the course
chat.

DELIVERABLE

Live, publicly accessible AI
web application

Phase 4 — Deploy & Ship

AI Agents & Deployment

From notebook to shareable web app



Concept + Code

Session 1

AI Agents & Tool Calling

- What are AI agents (vs chatbots)
- Tool/function calling explained
- Building a simple agent loop
- Agent use cases in IT operations



Hands-On Lab

Session 2

Deploy with Streamlit

- Streamlit: web UI in 10 lines of Python
- Deploy to Hugging Face Spaces (free)
- API key management & secrets
- Lab: Deploy your RAG app online

8

WEEK

Phase 4
Deploy & Ship

All Layers

HOMEWORK

Write a 1-paragraph LinkedIn post about what you built.

DELIVERABLE

Capstone project +
LinkedIn-ready portfolio

Phase 4 — Deploy & Ship

Capstone Project & Career Path

Demo day — show what you built



Project Build

Session 1

Capstone Project

- Personal AI project: choose your domain
- Build, polish, and document your app
- Peer review & feedback round
- Fix issues, improve quality



Demo & Review

Session 2

Demo Day & What's Next

- Live project demos (5 min each)
- AI Engineering career roles & paths
- Layer 3 deep-dive roadmap (optional)
- Certifications & continued learning

3-Layer Learning Architecture

Where each layer fits across the 8 weeks.

Layer 1 — Build Fast

Goal: Confidence + momentum

 **Weeks 1–8 (throughout)**

- ▶ Prompting & Prompt Engineering
- ▶ AI APIs (Anthropic, OpenAI)
- ▶ RAG Systems
- ▶ AI Agents & Workflows
- ▶ SaaS Deployment with Streamlit

Layer 2 — Understand Internals

Goal: Intuition without the math

 **Weeks 4–6 (deepens)**

- ▶ Embeddings & Vector Databases
- ▶ Chunking Strategies
- ▶ Transformers (conceptual)
- ▶ Context Windows & Tokens
- ▶ Hallucinations & Mitigation

Layer 3 — Deep Foundations

Goal: Optional advanced path

 **Post-course (self-paced)**

- ▶ Machine Learning & Fine-tuning
- ▶ Attention Mechanism
- ▶ Neural Architectures
- ▶ Optimization & Training
- ▶ Model Evaluation

Most professionals become highly productive AI engineers with Layers 1 & 2 alone.

By Week 8, you will have built:

A real, deployed AI application.

- ✦ A working RAG chatbot over your own documents
- ✦ A Streamlit web app deployed publicly online
- ✦ A personal AI portfolio you can show your employer
- ✦ The foundations to keep building independently

Build. Learn. Go deep — progressively.